OFFICIAL ARCHITECTURE & OPERATIONAL DOCUMENT

# Recruitment Lifecycles, End-to-End Workflows & User Persona Use Cases

Comprehensive functional specification, operational pipelines, RBAC permission models, and ethical AI safeguards.

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STATUS / RELEASE
v1.0 Production Certified

## 1. Executive Summary & Architectural Overview

**RecruitAI** is a next-generation Applicant Tracking System engineered to eliminate hiring bottlenecks while enforcing strict algorithmic fairness, privacy compliance (GDPR/Law N° 058/2021), and full decision explainability. Unlike opaque screening black-boxes, RecruitAI serves as an intelligent *decision-support co-pilot*: candidates are evaluated against objective job rubrics, with direct verbatim citations anchored to the candidate resume, while humans retain absolute control over shortlisting and hiring decisions.

## 2. The 8-Stage End-to-End Recruitment Lifecycle

The platform coordinates talent acquisition across an 8-stage automated and human-governed pipeline:

STAGE 1 · DEFINITION

### Requisition & Rubric Setup

Hiring Managers and Recruiters post vacancies, defining core requirements, preferred skills, minimum experience, and weighted evaluation criteria (e.g. Technical 40%, Leadership 20%, Domain 40%).

STAGE 2 · INTAKE

### Application Sourcing & Ingestion

Candidates apply via public job portals or recruiters ingest bulk resume archives (PDF/DOCX up to 5MB) with automated pre-flight format validation and virus/malware scanning.

STAGE 3 · PRIVACY & BIAS SHIELD

### PII Redaction & Blind Screening

The parser automatically detects and redacts personal identifiers (names, emails, phone numbers, addresses, graduation years, universities) to mitigate demographic proxy bias.

STAGE 4 · AI ANALYSIS

### Scoring & Explainable Citations

The AI computes overall match scores (0–100%) and extracts direct evidence quotes from the resume, mapping qualifications to rubric criteria with confidence flags for human review.

STAGE 5 · PIPELINE TRIAGE

### Deduplication & Quick-Filtering

Recruiters resolve potential duplicate submissions via fuzzy contact matching and triage candidates using 1-click filter chips (Needs Review, Top Matches, Required Failed).

STAGE 6 · EVALUATION

### Interviews & Structured Scorecards

Recruiters coordinate interview rounds; interviewers utilize AI-generated competency question guides and submit standardized 5-point scorecards with qualitative notes.

STAGE 7 · DECISION

### Hiring Decisions & Offer Dispatch

Hiring Managers make final selections. Overrides of AI recommendations require mandatory logged rationales before customized offer letters or gentle rejections are dispatched.

STAGE 8 · GOVERNANCE

### Tamper-Evident Audit & Telemetry

Every event is cryptographically sealed in a SHA-256 blockchain-style hash chain. Auditors verify demographic parity and disparate impact across recruitment cohorts.

**Core Architectural Guarantee:** The AI engine never autonomously rejects or accepts an applicant. All hiring outcomes require authenticated human intervention, reinforced by cryptographic accountability.

# 3. User Personas & Use Case Specifications

## 👤 Persona 1: Candidate (Job Applicant)

EXTERNAL JOB SEEKER

External users seeking employment. Restricted strictly to public career portals and their own personal application tracking dashboard.

> **UC-C1: Public Job Discovery & Application Intake**
> **Route:** `/jobs`, `/candidates/apply` · Search open roles, filter by department, review requirements, and submit a resume (PDF/DOCX ≤ 5MB) with client-side format and file-size validation.

> **UC-C2: Dual Portal Candidate Authentication**
> **Route:** `/auth/signin`, `/auth/signup` · Register/sign in via the dedicated "Work Applicant" portal tab with email verification and secure session tokens.

> **UC-C3: Stepped Application Progress Tracking**
> **Route:** `/dashboard/my-applications` · Monitor real-time status via a visual 4-step progress stepper (Submitted → Screening → Interviewing → Decision) with active status badges.

> **UC-C4: Data Subject Privacy & Export Rights**
> **Route:** `/privacy`, `/dashboard/profile` · Exercise GDPR / Law N° 058/2021 rights including requesting data export or executing the Right to Be Forgotten.

## 📋 Persona 2: Recruiter (Talent Acquisition Specialist)

PIPELINE OPERATIONAL LEAD

Talent acquisition professionals responsible for end-to-end pipeline execution, resume ingestion, candidate filtering, deduplication, and communications.

> **UC-R1: Job Requisition & Rubric Authoring**
> **Route:** `/jobs`, `/api/jobs` · Author job postings, establish required vs preferred skill matrices, and configure weighted rubric criteria.

> **UC-R2: Resume Ingestion & Blind Screening Control**
> **Route:** `/candidates`, `/api/candidates` · Upload batches of resumes, trigger AI parsing, and toggle blind screening to mask demographic identifiers.

> **UC-R3: Fast Triage via 1-Click Filter Chips**
> **Route:** `/candidates` · Rapidly filter large candidate volumes using preset chips: ⚠️ `Needs Review`, ⭐ `Top Matches`, ❌ `Required Failed`, and 🗓️ `Interviewing`.

> **UC-R4: Duplicate Candidate Detection & Resolution**
> **Route:** `/candidates/duplicates` · Inspect suspected duplicate profiles, view historical submission diffs, and merge duplicate submissions into an authoritative record.

> **UC-R5: Multi-Candidate Side-by-Side Comparison**
> **Route:** `/candidates/compare` · Compare 2 to 4 candidate qualifications side-by-side with sticky headers, criteria matrix scores, and radar overlays.

> **UC-R6: Candidate Stage Notifications & Talent Pooling**
> **Route:** `/api/candidates/[id]/interview`, `/api/jobs/[id]/talent-pool` · Issue automated branded email notifications and tag candidates for future talent pools.

## 👔 Persona 3: Hiring Manager (Department Head)

HIRING DECISION AUTHORITY

Departmental leadership focused on candidate profile deep-dives, interview coordination, scorecard consensus, and final hiring selections.

> **UC-HM1: Deep Evidence Verification & Auto-Scroll**
> **Route:** `/candidates/[id]` · Review AI match scores, inspect competency breakdowns, and click "Locate Evidence" to auto-scroll directly to verbatim quotes in the resume viewer.

> **UC-HM2: Interview Round Scheduling & Panel Assignment**
> **Route:** `/api/candidates/[id]/interview` · Configure multi-stage interview rounds (Technical, Cultural, Executive), assign interviewers, and generate calendar invites.

> **UC-HM3: Scorecard Consolidation & Consensus Review**
> **Route:** `/candidates/[id]` · Review completed panel scorecards, evaluate rating divergence, and review qualitative interviewer commentary.

> **UC-HM4: Human-in-the-Loop Override & Decision Issuance**
> **Route:** `/api/decisions`, `/api/candidates/[id]/offer` · Override AI recommendations with mandatory logged justifications and issue formal Offers or Rejections.

## 🎯 Persona 4: Interviewer (Panel / Technical Evaluator)

ASSESSMENT SPECIALIST

Engineers, team members, or domain specialists assigned to conduct interviews. Granted strictly scoped evaluation capabilities with no final decision rights.

> **UC-I1: Scoped Candidate Profile Review**
> **Route:** `/candidates/[id]` · Access candidate resume highlights and qualification summaries strictly for assigned interview rounds.

> **UC-I2: AI-Generated Interview Question Guides**
> **Route:** `/candidates/[id]` · Access AI-curated behavioral and technical questions targeting specific skill gaps or unverified claims in the resume.

> **UC-I3: Standardized Scorecard Submission**
> **Route:** `/api/candidates/[id]/scorecard` · Complete structured 5-dimension scorecards (1–5 scale), verify mandatory technical skills, and submit final recommendations (Strong Yes, Yes, No, Strong No).

## ⚖️ Persona 5: Compliance Auditor (Legal / Ethics / DEI Officer)

GOVERNANCE & INDEPENDENT OVERSIGHT

Legal, ethical, and compliance officers responsible for algorithmic accountability, bias detection, data protection, and tamper verification.

> **UC-CA1: Cryptographic Audit Explorer**
> **Route:** `/audit` · Filter and review all system actions by actor, action type, target entity, timestamp, and IP address.

> **UC-CA2: Tamper-Evident Hash Chain Verification**
> **Route:** `npm run verify-audit-chain` · Execute cryptographic verification proving that every audit entry is linked via unbroken SHA-256 hashes: `Hash(n) = SHA256(Hash(n-1) + Payload)`.

> **UC-CA3: Algorithmic Fairness & Disparate Impact Auditing**
> **Route:** `/dashboard/analytics` · Inspect demographic parity metrics, verifying that selection rates satisfy the EEOC 4/5ths (80%) rule across demographic proxies.

> **UC-CA4: Human Override Surveillance**
> **Route:** `/audit` · Review all instances where recruiters or hiring managers deviated from AI recommendations to detect human bias.

## 🛡️ Persona 6: Workspace Administrator (System Owner / IT Lead)

FULL SYSTEM GOVERNANCE

Workspace administrators responsible for platform infrastructure, security enforcement, team membership, and operational telemetry.

> **UC-A1: Team RBAC & Role Management**
> **Route:** `/dashboard/team` · Invite team members, reassign roles, configure granular permission overrides, suspend accounts, and safely delete members with confirmation modals.

> **UC-A2: Multi-Factor Authentication (TOTP 2FA) Enforcement**
> **Route:** `/dashboard/profile`, `/api/auth/mfa/*` · Configure mandatory 2FA enrollment via TOTP authenticator apps or 6-digit email OTP challenges.

> **UC-A3: First-Login Password Reset Protocol**
> **Route:** `/auth/first-login` · Force newly invited staff to establish personal passwords (≥8 characters, bcrypt cost 12) prior to accessing platform workspaces.

> **UC-A4: Real-Time System Health Diagnostics**
> **Route:** `/dashboard/system-health` · Monitor database query latency, Redis queue workers, storage providers (Local/S3), and anti-malware scanner status.

> **UC-A5: Email Delivery Monitoring & Dead-Letter Queue Retries**
> **Route:** `/dashboard/email-delivery` · Review transactional email logs (Gmail/Brevo/Resend) and trigger dead-letter delivery retries for failed dispatches.

## 4. Formal RBAC Permission Matrix

The platform implements strict permission-gated authorization defined in `src/lib/roleAccess.ts`:

| PERMISSION IDENTIFIER | CAPABILITY / ACTION | ADMIN | RECRUITER | HIRING MGR | INTERVIEWER | AUDITOR | CANDIDATE |
|---|---|---|---|---|---|---|---|
| `jobs:manage` | Create, edit, publish & archive job postings | ✓ | ✓ | ✓ | × | × | × |
| `jobs:rubrics` | Configure scoring criteria & weighting weights | ✓ | ✓ | ✓ | × | × | × |
| `candidates:review` | View candidate profiles & AI evidence matches | ✓ | ✓ | ✓ | ✓ | ✓ | × |
| `candidates:override-scores` | Manually adjust scores with logged justification | ✓ | ✓ | ✓ | × | × | × |
| `candidates:merge-duplicates` | Merge suspected duplicate candidate records | ✓ | ✓ | × | × | × | × |
| `interviews:schedule` | Schedule interview rounds & dispatch calendar invites | ✓ | ✓ | ✓ | × | × | × |
| `interviews:scorecards:submit` | Submit structured interview evaluation scorecards | ✓ | ✓ | × | ✓ | × | × |
| `candidates:decide` | Execute final hiring decision (Offer / Reject) | ✓ | ✓ | ✓ | × | × | × |
| `candidates:offer` | Generate and dispatch formal job | ✓ | ✓ | × | × | × | × |

| PERMISSION IDENTIFIER | CAPABILITY / ACTION | ADMIN | RECRUITER | HIRING MGR | INTERVIEWER | AUDITOR | CANDIDATE |
|---|---|---|---|---|---|---|---|
| | offer documents | | | | | | |
| `compliance:audit-log` | Inspect immutable cryptographic audit trail | ✓ | × | × | × | ✓ | × |
| `ops:email-delivery:view` | Inspect email dead-letter queue & retry deliveries | ✓ | × | × | × | × | × |
| `team:manage` | Invite staff, reassign roles, configure RBAC | ✓ | × | × | × | × | × |
| `applicant:portal` | Submit applications and track stage progress | × | × | × | × | × | ✓ |

## 5. Security, Ethical AI & Compliance Guarantees

### 1. Verifiable Evidence Anchoring

Every AI score links directly to extracted text from the applicant's resume. Clicking "Locate Evidence" scrolls the viewer directly to highlighted sentences, preventing hallucinated evaluations.

### 2. Cryptographic Tamper-Evidence

Audit logs calculate SHA-256 hashes linking every row to the preceding log record. Any manual database alteration immediately breaks the mathematical chain, exposing unauthorized modifications.

### 3. Algorithmic Fairness & 4/5ths Rule

The scoring engine enforces demographic parity checks. Statistical monitors flag selection differentials exceeding the 80% threshold across gender, age, and institutional proxies.

### 4. Defense-in-Depth Authentication

Enterprise 2FA incorporates TOTP authenticator apps, single-use 6-digit email OTP challenges, first-login password enforcement, and strict bcrypt hashing with cost factor 12.

**Verification & Compliance Certified:** Built and verified under CODAFRIQA Intern Project Specification v1.0. All 232 test cases passed; cryptographic audit chain validated 100% intact.